

Web Application Security Home Lab using Kali Linux and OWASP Juice Shop

Objective

- The objective of this project was to build a local web application security lab and understand how web applications communicate using HTTP requests, APIs, and authentication mechanisms.

Tools Used:

- Kali Linux
- OWASP Juice Shop
- Burp Suite

Steps Performed

1. Kali Linux Installed on Virtual Box
2. Docker Installed and configured.
3. OWASP Juice Shop running locally on port 3000
4. Set up Burp Suite as a web proxy
5. Intercepting and observing HTTP requests
6. API calls and login request analysis
7. Send requests to Burp Repeater for manual testing

Observation

During testing, OWASP Juice Shop was successfully deployed in a controlled virtual lab environment with Kali Linux and Docker. We set up Burp Suite as an intercepting proxy that enabled us to capture and analyse the HTTP requests and responses between the browser and web application.

Analysis of the network traffic revealed that user interactions with the application resulted in HTTP requests to various API endpoints. Both GET and POST methods were seen. GET requests were used to retrieve application data and POST requests were used to submit user input such as login credentials.

Using the interception functionality of Burp Suite, we could pause, inspect and forward requests before they reached the target application. It gave an insight of how modern web applications process user actions and communicate to backend server.